

Curriculum Vitae: Olivia Lewis

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY

Master of City Planning student. Proposed concentration: transportation planning

Berkeley, CA

August 2026- May 2028

TUFTS UNIVERSITY

Bachelor of Arts in Community Health, Spanish Literature

Medford, MA

August 2019-May 2023

- Major Advisor, Community Health: John Fu, MD PhD
- Major Advisor, Spanish Literature: Maria Ester Rincon Calero, A.B.D. PhD
- GPA: 3.92 (Summa Cum Laude)

PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE

Study Abroad Semester: advanced Spanish grammar, demography, and intercultural health

Santiago, Chile

February-July 2022

RESEARCH EXPERIENCE

TUFTS UNIVERSITY SCHOOL OF MEDICINE

Boston, MA

Research Assistant, Dr. Thomas Stopka's GIS and Spatial Epidemiology Research Lab

August 2022- June 2026

Research staff member in the Department of Public Health and Community Medicine. Projects listed below:

Predict to Prevent: Dynamic Spatiotemporal Analyses of Opioid Overdose to Guide Pre-Emptive Public Health Responses.

MPIs: Drs. Thomas Stopka and Cici Bauer. 2025-2026. Secondary data analysis to predict opioid overdoses in Massachusetts by employing Bayesian spatiotemporal models, informing pre-emptive public health responses.

- Plan community advisory board meeting by coordinating logistics, facilitating community engagement process with stakeholders with diverse experiences, expertise, and perspectives in the field of overdose prevention
- Coordinate logistics for P2P study, drafting agendas, scheduling meetings, and taking detailed meeting notes

Xylazine in Lowell: Community-Engaged Spatial Analysis and Harm Reduction.

PI: Dr. Jack Cordes. 2024-2026. Cross-sectional study aiming to quantify associations between xylazine exposure and violence in Lowell, MA.

- Perform street outreach and survey people who use drugs on xylazine exposure and experiences of violence
- Work in ArcGIS Pro to generate choropleth maps and hotspot cluster analyses, identifying areas of concentrated substance and xylazine-related violence in Lowell to inform local outreach efforts of high-priority areas

Examination of Xylazine Exposure and the Risk of Skin and Soft Tissue Infections in People Who Inject Drugs.

MPIs: Drs. Shikhar Shrestha and Thomas Stopka. 2024-2026. Prospective cohort study to assess whether exposure to xylazine is associated with the development of soft skin/tissue infections in a cohort of people who inject drugs

- Conduct in-depth interviews with participants to learn more about their experiences with xylazine and wounds
- Perform data entry and quality control to ensure accuracy of results

Multilevel Physical Activity Intervention for Low Income Public Housing Residents.

PI: Dr. Lisa Quintilliani, 2024-2026. Randomized controlled trial evaluating a community health worker behavioral counseling and text messaging intervention designed for weight loss among public housing residents

- Create choropleth maps of crime rates to recruit participants living in areas of high crime for in-depth interviews
- Use R to determine travel times from participants' residences to grocery stores where they food shop

Geographic Access to Community-Based Opioid Treatment Programs (OTPs).

PI: Dr. Thomas Stopka. 2023-2025. Secondary data analysis project analyzing geographic access to OTPs in Massachusetts to inform targeting and funding efforts for the Massachusetts Department of Public Health Bureau of Substance Addiction Services (BSAS)

- Mapped geographic access to OTPs by creating walk and drive-time buffers in ArcGIS Pro, overlaying decedent data to identify areas with need for methadone treatment
- Collaborated with co-authors to write summary reports, informing placement of new OTPs based on areas with low geographic access to methadone and fatal opioid overdose/ opioid-related incident locations
- Performed location-allocation analyses in a GIS to identify priority areas for placing new OTPs based on existing access to treatment and overdose locations in targeted regions of Massachusetts and on a statewide level

- As part of a community-engaged process, presented findings to BSAS and OTP providers to contextualize our results and inform stakeholders of areas with need for expansion of treatment
- Wrote documents to submit in eIRB for study approval

Jurisdiction-Level Substance Use-Related Vulnerability Analysis. MPIs: Drs. Thomas Stopka and Shikhar Shrestha. 2023-2025. Secondary data analysis project assessing opioid, alcohol, and stimulant-related vulnerability across Idaho

- Performed risk mapping of factors tied to substance use and treatment access to prioritize areas for intervention
- Worked in Python and STATA to design statistical models and calculate substance-related vulnerability indices
- Wrote report of findings to inform the Idaho Department of Health and Welfare of areas to target for interventions

HEALing Communities Study (HCS) in Massachusetts. PI: Dr. Thomas Stopka. 2022-2025. Worked on a GIS team, conducting spatial data analysis to identify high-need areas for opioid use disorder treatment

- Mapped hotspot cluster analyses of opioid-related overdose deaths and walk and drive-time access to treatment to inform study communities of high-need areas for expanding access to harm reduction and treatment resources
- Assessed the impact of opening three new OTPs on local changes in geographic access to methadone for communities with high social vulnerability to inform future strategic placement of new treatment centers

Syndromic Surveillance to Prevent Overdose in Lowell, MA: Data to Action and Evaluation (STOP-OD Lowell).

PI: Dr. Shikhar Shrestha. 2023-2024. Project aiming to decrease the time to respond to opioid-related incident (ORI) risk spikes by improving surveillance and communication systems, reducing opioid overdose deaths in Lowell, MA.

- Assisted in the creation of an automatically updating dashboard showing relative ORI risk by neighborhood
- Performed primary data collection by surveying people who use drugs in Lowell

PUBLICATIONS

Peer-Reviewed: Accepted and Published

- **Lewis, O.,** Srinivasan, S., Shrestha, S., Harris, D. R., Barboza-Salerno, G., Rock, P., ... Stopka, T. J. (2026). Enhancing Access to Methadone Treatment in Three HEALing Communities Study Sites: A Spatial and Geostatistical Study. *Substance Use & Misuse*, 1–12. <https://doi.org/10.1080/10826084.2026.2679711>
- Soares, W. E., **Lewis, O.,** Shrestha, S., Stopka, T. J., & Friedmann, P. D. (2026). “When” matters: Differences in fatal opioid-related overdose events in Massachusetts by time of day, 2019–2023. *Drug and Alcohol Dependence*, 283, 113141. <https://doi.org/10.1016/j.drugalcdep.2026.113141>
- Shrestha, S., **Lewis, O.,** Harris, D., Rock, P., Silwal, A., Srinivasan, S., & Stopka, T. J. (2025). A multi-method spatial examination of factors associated with changes in geographic accessibility to buprenorphine providers in HEALing communities study states Kentucky, Massachusetts, and Ohio. *Preventive Medicine Reports*, 53, 103045. <https://doi.org/10.1016/j.pmedr.2025.103045>
- Srinivasan, S., Shrestha, S., Harris, D. R., **Lewis, O.,** Rock, P., Silwal, A., Pustz, J., Oh, S., Barboza-Salerno, G., & Stopka, T. J. (2025). Employment industry and opioid overdose risk: A pre- and post-COVID-19 comparison in Kentucky and Massachusetts 2018–2021. *Spatial and Spatio-Temporal Epidemiology*, 52, 100701. <https://doi.org/10.1016/j.sste.2024.100701>
- O'Reilly, M. R., Stopka, T. J., Bayly, R., **Lewis, O.,** Shrestha, S., Cordes, J., Walley, A. Y., & Srinivasan, S. (2025). Public transit and methadone – Spatial analyses of opioid treatment program access in Greater Boston, 2020–2022. *Preventive Medicine Reports*, 60, 103317. <https://doi.org/10.1016/j.pmedr.2025.103317>
- Parbs, J. R., Srinivasan, S., Pustz, J., Bayly, R., Shrestha, S., **Lewis, O.,** Kimmel, S., Meehan, T., Babakhanlou-Chase, H., & Stopka, T. J. (2024). The optimization of harm reduction services in Massachusetts through the use of GIS: Location-allocation analyses, 2019–2021. *Preventive Medicine*, 186, 108088. <https://doi.org/10.1016/j.ypmed.2024.108088>
- Harris, D. R., Shrestha, S., Rock, P., Silwal, A., Barboza-Salerno, G., **Lewis, O.,** Srinivasan, S., & Stopka, T. J. (2025). The impact of buprenorphine prescriber data on geospatial access to treatment in HEALing Communities Study communities, 2022. *Journal of Substance Use and Addiction Treatment*, 171, 209625. <https://doi.org/10.1016/j.josat.2025.209625>

Government Reports

- **Lewis O,** Shrestha S, and Stopka TJ. Opioid-Related Overdose Vulnerability in Idaho: An Epidemiological Assessment. July 26, 2024. Department of Public Health and Community Medicine. Tufts University School of Medicine. PDF available upon request.

- **Lewis O.**, Shrestha S., and Stopka TJ. Alcohol-Related Vulnerability in Idaho: An Epidemiological Assessment. February 20, 2025. Department of Public Health and Community Medicine. Tufts University School of Medicine. PDF available upon request.
- **Lewis O.**, Shrestha S., and Stopka TJ. Stimulant-Related Vulnerability in Idaho: An Epidemiological Assessment. April 7, 2025. Department of Public Health and Community Medicine. Tufts University School of Medicine. PDF available upon request.
- Stopka TJ, **Lewis O.**, Cordes J, Bayly R, Srinivasan S, O'Reilly M, Shrestha S. Geographic Access to Community-Based Opioid Treatment Programs in Massachusetts. May 23, 2024. Department of Public Health and Community Medicine. Tufts University School of Medicine. (<https://www.mass.gov/doc/geographic-access-to-community-based-opioid-treatment-programs-in-massachusetts/download>)
- **Lewis O.**, Srinivasan S., Cordes J., Bayly R., Shrestha S., O'Reilly M., and Stopka TJ. (2025). Geographic Access to Opioid Treatment Programs (OTPs), Boston, Massachusetts, 2023-2024, Using Location-Allocation Analysis to Identify Optimal Placement for New OTPs. Department of Public Health and Community Medicine. Tufts University School of Medicine. PDF available upon request.
- **Lewis O.**, Srinivasan S., Cordes J., Bayly R., Shrestha S., O'Reilly M., and Stopka TJ. (2025). Geographic Access to Opioid Treatment Programs (OTPs), Western Massachusetts, Massachusetts, 2023-2024, Using Location-Allocation Analysis to Identify Optimal Placement for New OTPs. Department of Public Health and Community Medicine. Tufts University School of Medicine. PDF available upon request.

CONFERENCE ABSTRACTS & PRESENTATIONS

- Shrestha S, **Lewis O.**, Kelley J, Stopka TJ. "Using a Modified Hawkes Process to Detect Spikes in Opioid-Related Incidents Using Emergency Medical Services Data, 2016-2022." *American Public Health Association Annual Meeting*. 2024 poster.

ADDITIONAL INVITED PRESENTATIONS

- Lewis O. "Attribute Tables and Queries." Guest Lecture, PH-262: Introduction to GIS for Public Health. *Tufts University School of Medicine, Department of Public Health and Community Medicine*. March 4, 2025.

OTHER EXPERIENCE

Tufts University School of Medicine, Boston, MA

Spring & Fall 2025

Teaching Assistant for PH-262, Introduction to GIS for Public Health, 5 hours/week

- Grade assignments to provide students with feedback and improve their mapping skills
- Hold weekly office hours to troubleshoot software issues and provide constructive criticism on projects
- Students reported "excellent" guest lecture and helpful feedback from TA for improving spatial mapping skills

AWARDS & HONORS

- Tufts University Dean's List: 2019-2023: For maintaining a GPA of 3.4 or higher
- Tufts University Summa Cum Laude: For graduating with a GPA of 3.8 or above

SKILLS & RELEVANT COURSEWORK

Select Relevant Coursework: Computer Science for All; Introduction to GIS; Advanced GIS; Introduction to Statistics; Spatial Statistics; Applied Quantitative Research Methods; Planning for Low-Impact Development

Technical Skills: ArcGIS products, Python, R, STATA, SQL, Qualtrics, Zotero, eIRB, CITI certified

Soft Skills: Leadership, collaboration, communication, organization, problem-solving, detail-oriented

Languages: English (Native), Spanish (Advanced skills in speaking, listening, reading, and writing)